

Prog 5/6

Introduction

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Controlling a stepper

- Programming a stepper motor
- The ESE dilemma
- The Prog 5/6 course
- Programming a stepper motor better

Controlling a stepper

- What do we need?
- How do we keep track of what we need?

Controlling a stepper

See code example `stepper_code_example_1a.ino`

- What are the pro's of this approach?
- What are the con's?

Controlling a stepper

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Controlling a stepper

- What are the pro's of this approach?

Another approach (MIC2-style ;)

- Directly program all registers

```
#include <avr/io.h>

void setup() {
    // D2 (PD2)=STEP, D3 (PD3)=DIR, D4 (PD4)=ENA as outputs
    DDRD |= (1<<2) | (1<<3) | (1<<4);

    // Enable driver (assumes ENA active-LOW), STEP=LOW, DIR=LOW
    PORTD &= ~(1<<4);
    PORTD &= ~((1<<2) | (1<<3));

    // Timer1: normal mode, prescaler = 8 (16 MHz / 8 = 2 MHz → 0.5 μs/tick)
    TCCR1A = 0;
    TCCR1B = (1<<CS11);
    // No Timer1 interrupts used
    TIMSK1 = 0;
}
```

```

void loop() {
  // ---- Clockwise: DIR HIGH, settle ~20  $\mu$ s (40 ticks) ----
  PORTD |= (1<<3);
  uint16_t t = TCNT1;
  while ((uint16_t)(TCNT1 - t) < 40) { } // ~20  $\mu$ s

  // 1 rev @ 16 $\times$  microstep = 3200 pulses
  for (unsigned int i = 0; i < 3200; i++) {
    PORTD |= (1<<2);      // STEP HIGH
    t = TCNT1; while ((uint16_t)(TCNT1 - t) < 20) { } // ~10  $\mu$ s HIGH
    PORTD &= ~(1<<2);     // STEP LOW
    t = TCNT1; while ((uint16_t)(TCNT1 - t) < 20) { } // ~10  $\mu$ s LOW
  }

  // ~500 ms pause using Timer1: 25 ms (50,000 ticks)  $\times$  20
  for (uint8_t k = 0; k < 20; k++) {
    t = TCNT1; while ((uint16_t)(TCNT1 - t) < 50000) { } // ~25 ms
  }

  // ---- Counterclockwise: DIR LOW, settle ~20  $\mu$ s ----
  ...

```

```
void loop() {  
    // ---- Clockwise: DIR HIGH, settle ~20  $\mu$ s (40 ticks) ----  
    ...  
  
    // ---- Counterclockwise: DIR LOW, settle ~20  $\mu$ s ----  
    PORTD &= ~(1<<3);  
    t = TCNT1;  
    while ((uint16_t)(TCNT1 - t) < 40) { } // ~20  $\mu$ s  
  
    // 1 rev back  
    for (unsigned int i = 0; i < 3200; i++) {  
        PORTD |= (1<<2);  
        t = TCNT1; while ((uint16_t)(TCNT1 - t) < 20) { }  
        PORTD &= ~(1<<2);  
        t = TCNT1; while ((uint16_t)(TCNT1 - t) < 20) { }  
    }  
  
    // ~500 ms pause again  
    for (uint8_t k = 0; k < 20; k++) {  
        t = TCNT1; while ((uint16_t)(TCNT1 - t) < 50000) { }  
    }  
}
```

Controlling a stepper

- What are the pro's of this approach?
- What are the con's?

Controlling a stepper

- What are the pro's of this approach?

The ESE-dilemma

- How do we program prototypes?

The ESE-dilemma

- How do we program prototypes?
- How do we program pacemakers?

The ESE-dilemma

- To summarize the first part of this introduction:
 - No matter what they tell you:
 - Embedded Software Engineering is:
 - The most vital subject of ESE
 - The most difficult subject of ESE
- People WILL die if we make a mess (more on that in MIC6 ;))
- (If you don't actively participate you won't get buck for you money...)

The ESE-dilemma

- We work in GitHub:
 - <https://github.com/AEAEmbedded/Course202526.git>
 - <https://classroom.github.com/classrooms/147585071-aeaembedded-course202526>
- Send me your GitHub username so I can link it to your account

The ESE-dilemma

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The ESE-dilemma

- Why ESE is really engineering:
 - Trade-off between readability and functionality
 - Efficient code might not be as readable
 - Adding abstractions will cost overhead (memory, performance)
 - Even well designed code is a trade-off between memory and processor intensiveness

Let's review `stepper_code_example_1a.ino`

Programming in MIC2 style

Disclaimer: Of course in MIC2 we wanted to show you how microcontrollers work under the hood. Different didactical goal for good reason!

The ESE-dilemma

Readability

- Is readable code always reliable?

The ESE-dilemma

Readability

- Is readable code always reliable?
- Is reliable code always readable?

The ESE-dilemma

Readability

- Is readable code always reliable?
- Is reliable code always readable?
- What is readable code?

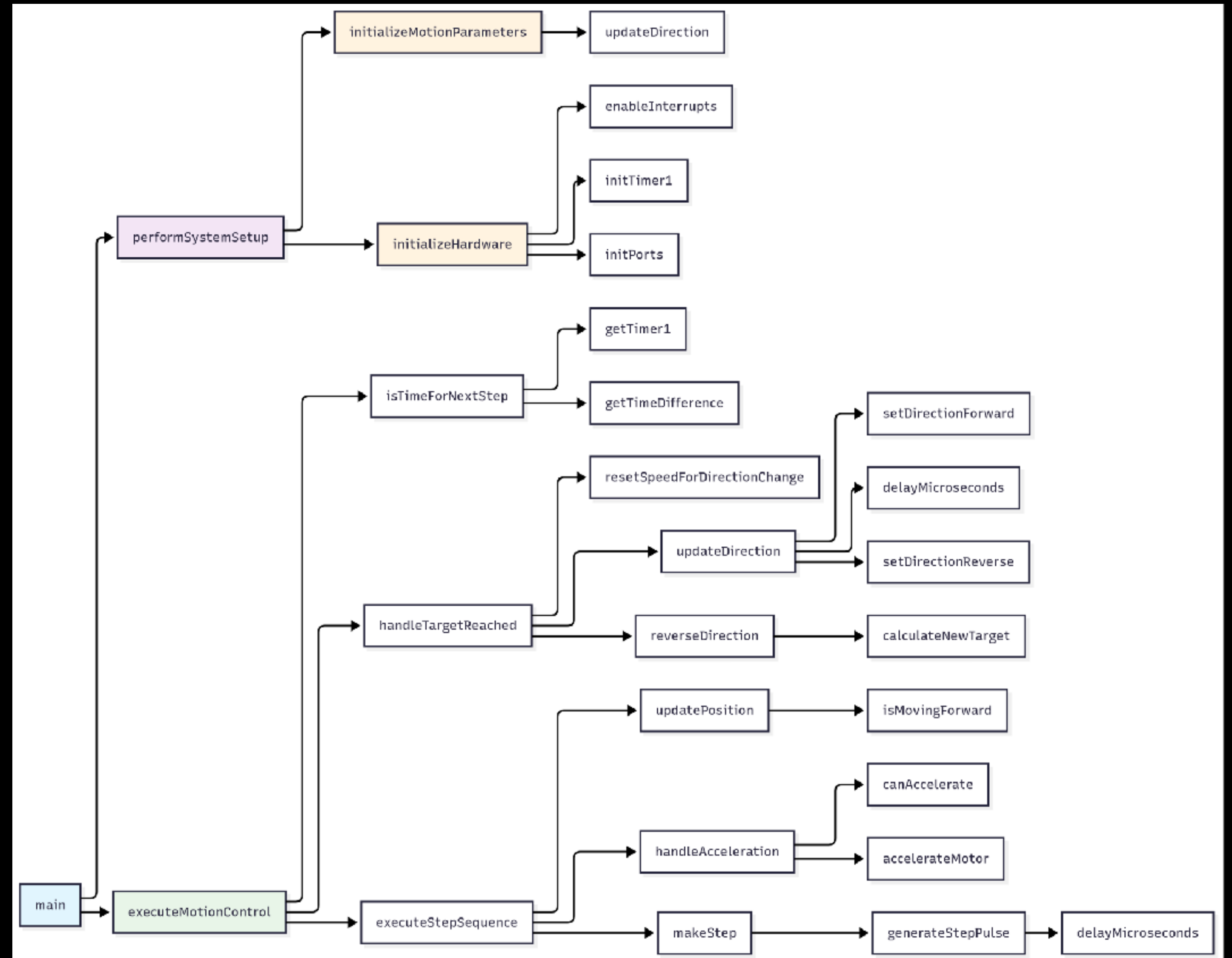
The ESE-dilemma

Readability

- Is readable code always reliable?
 - Is reliable code always readable?
 - What is readable code?
-
- WARNING: Try to avoid hacking (e.g. overruling the compiler, unsafe casting, etc)

Let's review stepper_code_example_1b.ino

Programming in MIC2 style
but now with methods



The ESE-dilemma

Readability

- Is the code structured well?
- Why is the code (not) readable?
- What code smells did you detect?
- Under which circumstance would this code be acceptable (at least partially)

Structured programming

- Define the properties you need
- Define the behavior you need

Structured programming

Properties

- Variables
- Constants (always preferred)
- Static vs non-static

Structured programming

Properties - Variables

- Variables
 - Variables are storage locations that hold data values that can change during program execution. They have:
 - A name (identifier) to reference them
 - A type (in typed languages) that determines what kind of data they can store
 - A value that can be modified (i.e. is mutable)

Structured programming

Properties - Constants

- Constants (Always Preferred)
 - Constants are immutable values that cannot be changed after initialization. They provide:
 - Immutability - prevents accidental modification
 - Performance benefits - compiler optimizations
 - Code clarity - signals intent that value shouldn't change
 - Thread safety - no risk of concurrent modification

Structured programming

Properties

- Static vs Non-Static
 - Static Members
 - Belong to the class itself, not individual instances
 - Shared across all instances of the class
 - Accessed using the class name directly
 - Initialized once when the class is first loaded
 - Non-Static (Instance) Members
 - Belong to specific instances of the class
 - Each object has its own copy
 - Must be accessed through an object instance
 - Initialized each time an object is created

Structured programming

Behavior

- Methods
 - void type:
 - By default: performs some action (in an embedded context this might as well be an action that changes the physical environment i.e. actuator)
 - Warning: try to avoid side effects (i.e. methods that modify properties without explicitly stating this in the method name).
- Methods
 - non-void type:
 - By default: modifies some properties and/or state

Let's review `stepper_code_example_1c.ino`

The ESE-dilemma

Readability

- Is the code structured well?
- Why is the code (not) readable?
- What code smells did you detect?

The ESE-dilemma

Resource management

The screenshot shows the Arduino IDE 2.3.5 interface. The top toolbar includes icons for checking, uploading, and debugging, along with a dropdown menu for the board, currently set to 'Arduino Uno'. The main editor displays the file 'stepper_code_example_1e.ino' with C++ code. The code includes a function `handleLimitReached` and a `void handleBackoff` function. The `handleBackoff` function contains a `long` variable `distanceBacked` and several `if` statements. The bottom panel shows the 'Output' tab with the following text:

```
Sketch uses 10616 bytes (32%) of program storage space. Maximum is 32256 bytes.
Global variables use 2332 bytes (113%) of dynamic memory, leaving -284 bytes for local variables. Maximum is 2048 bytes.
Not enough memory; see https://support.arduino.cc/hc/en-us/articles/360013825179 for tips on reducing your footprint.
data section exceeds available space in board

Compilation error: data section exceeds available space in board
```

A notification box at the bottom right displays the error message: 'Compilation error: data section exceeds available space in board' with a 'COPY ERROR MESSAGES' button. The status bar at the bottom indicates 'Ln 823, Col 14' and 'Arduino Uno on /dev/cu.usbmodem111401'.

```
stepper_code_example_1e.ino
609     if (triggeredLimit) {
610         handleLimitReached(triggeredLimit);
611     } else {
612         motor_.step();
613         profile_.accelerate();
614     }
615 }
616 }
617
618 void handleBackoff() {
619     motor_.step();
620
621     long distanceBacked = abs(motor_.getPosition() - backoffStart_);
622     if (distanceBacked >= backoffTarget_) {
623         motionState_ = RUNNING;
624         Serial.println("Backoff complete, resuming normal operation");
625         profile_.resetForDirectionChange();
626         limitManager_.resetAll();
627     }
628 }
629 }
```

Output Serial Monitor

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✖ Compilation error: data section exceeds available space in board

COPY ERROR MESSAGES

Ln 823, Col 14 Arduino Uno on /dev/cu.usbmodem111401 3

Course

- Send me your student number + GitHub account name
- I will only use Brightspace for practical information as this course tries to come as close to real development as possible.
- Tips to get the most from this course:
 - Come prepared
 - Practice makes perfect!
 - You can never program too much
 - Vibe-coding: typically avoid, however useful for diagramming, documenting and refactoring, code-reviewing, use with caution

One more thing...

- Check code against code standards (e.g. MISRA)
- Use cppcheck and other static code analyzers
- Carefully check:
 - Your code
 - Your colleagues' (pair programming, code reviews)
 - Your teachers' (including me)
 - You can always
- TIP: Make use of git pull request and issues